

BASIL C K

AI Engineer | Machine Learning · LLM Systems · Computer Vision · RAG Pipelines

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PROFILE

AI Engineer Intern building production AI systems spanning Computer Vision, LLM integration, RAG pipelines, and AI workflow orchestration. Currently developing an AI model evaluation and orchestration platform for testing multi-provider LLMs while delivering Computer Vision and Generative AI solutions at Difinity Digital. Published research at IEEE ICNPCV 2026 on hybrid deep learning frameworks. Passionate about scalable AI infrastructure — from model experimentation to deployment-ready intelligent systems.

TECHNICAL SKILLS

AI & Machine Learning: Computer Vision, Deep Learning, CNNs, YOLO, LLM Integration, RAG (Retrieval-Augmented Generation), Prompt Engineering, Model Evaluation & Benchmarking, AI Workflow Orchestration, Conversational AI, AI Agent Development

LLM & Generative AI: Google Gemini AI, OpenAI-Compatible APIs, SSE Streaming, Multi-Model Routing, Vector Embeddings (pgvector), Semantic Search, AI Memory Systems, Agentic AI Pipelines

Frameworks & Libraries: TensorFlow/Keras, PyTorch, OpenCV, FastAPI, Flask, SQLAlchemy, Pydantic, NumPy, Pandas

Infrastructure & Tools: Python, SQL, Docker, Docker Compose, Git, Linux, PostgreSQL, GitHub Actions CI/CD, VS Code, Jupyter Notebook

Cloud & Platforms: Oracle Cloud Infrastructure (OCI AI Foundations), AWS Fundamentals, Google AI Essentials

EXPERIENCE

AI Engineer Intern — Difinity Digital

06/2026 – Present

Kochi, Kerala

- Developing AI-powered applications using Python, FastAPI, and REST APIs for real-world client deployments.
- Building Computer Vision solutions and Generative AI integrations including LLM-based workflow automation.
- Designing production-ready backend services and AI inference pipelines while collaborating with cross-functional engineering teams.

PROJECTS

OpenMind AI Platform — AI Model Evaluation & LLM Orchestration Platform **ONGOING**

07/2026 – Present

Python | FastAPI | Gemini AI | LLM APIs | RAG | pgvector | SQLAlchemy | Docker | PostgreSQL

- Building an AI model evaluation platform to test, compare, and benchmark LLMs across API-driven providers (Gemini, GPT-4o) with planned expansion to locally-hosted open-weight models (Llama 3, DeepSeek R1) via Ollama/vLLM.
- Designed a multi-session AI playground with configurable inference parameters (temperature, top_p, max_tokens, system prompts) and real-time SSE streaming for interactive model response evaluation.
- Implementing a RAG pipeline with PDF/Markdown document ingestion, text chunking, and vector embedding infrastructure (pgvector) for semantic knowledge retrieval over custom documents.
- Architecting a graph-based multi-tier memory system (Conversation, Semantic, Long-Term) to enable persistent AI context across sessions — a core building block for agentic AI applications.
- Developing a multi-step AI workflow orchestration engine supporting LLM calls, conditional branching, API integration, memory retrieval, and human-in-the-loop approval steps.
- Engineered an OpenAI-compatible API gateway with Bearer token authentication, enabling standardized model inference routing across 40+ API endpoints with structured logging and monitoring.

Hybrid CNN–YOLO Framework for Car Damage Severity Classification and Localization

2025 – 2026

Python | TensorFlow/Keras | YOLO | OpenCV | FastAPI | Computer Vision

- Developed an end-to-end hybrid deep learning framework integrating YOLO-based object detection with CNN-based severity classification for automated vehicle damage assessment.
- Built the complete ML pipeline including dataset preparation, model training, hyperparameter tuning, evaluation metrics, and a FastAPI inference backend for real-time predictions.
- Presented the research at the 2026 IEEE International Conference on Natural Language Processing and Computer Vision (ICNPCV), Bengaluru, India.

Vocalix — AI Voice Assistant

2025

Python | Speech Recognition | LLM APIs | Automation Libraries

- Built a voice-controlled AI assistant combining speech recognition with LLM inference for natural language task automation and system-level command execution.
- Engineered a real-time speech-to-LLM-to-action pipeline with modular command processing and dynamic response generation.

PUBLICATIONS

A Hybrid CNN–YOLO Framework for Car Damage Severity Classification and Localization

2026 IEEE International Conference on Natural Language Processing and Computer Vision (ICNPCV), Bengaluru, India

- Published and presented peer-reviewed research on a hybrid deep learning framework combining YOLO object detection with CNN classification for automated vehicle damage assessment.

EDUCATION

B.Tech in Artificial Intelligence and Data Science

2022 – 2026

JCT College of Engineering and Technology, Anna University

CGPA: 8.47 / 10

CERTIFICATIONS

- Oracle Cloud Infrastructure 2023 — AI Foundations Associate
- Google AI Essentials
- IBM Data Science — Selected Coursework (Python, Data Analysis, Machine Learning)

LANGUAGES

English · Malayalam · Tamil